Operations and recovery

Supported deployment

64-bit Windows 10/11, local writable disk, one application process per workspace. All application-owned data stays in that workspace. Do not put an open WAL database on a network filesystem or let a synchronization utility copy its files independently. Public internet exposure, authentication, TLS termination and multi-instance operation are outside this local deployment.

Default ports: API/UI 8000 and Prometheus 9090, both on `127.0.0.1`. Startup refuses occupied ports; it never terminates another application.

The complete Git checkout includes the runtime and database through Git LFS. Run `git lfs pull` before startup if the binaries are pointer files. The portable Release ZIP contains the actual files and needs no Git LFS client. Live collection modifies the tracked database snapshot; stop the app before committing or switching database revisions. Logs and transient state are not tracked.

Normal operation

```
.\Start-CleanAir.cmd
```

The launcher verifies local prerequisites, starts bundled Prometheus and its local storage, waits for monitoring readiness, then starts the app and one collector thread. A workspace file lock prevents a second application or manual writer. The browser opens only after the application's live endpoint responds.

Collection uses bounded timeouts, retries for transport/429/5xx failures and ETag conditional downloads. Official metadata is refreshed daily. Successfully imported files and their aggregates commit together with their checkpoints. An interrupted file is replayed safely; completed files remain committed.

The initial import must complete before comparisons are enabled. Once history exists, collection failures do not make cached historical queries unavailable. Provisional values, stale source data and failed updates are disclosed.

Current and historical searches

The new default view is official Bulgarian station detail with PM10. Blank dates compare the latest common reported hour, within six hours by default; an older database does not receive a stale "current" winner. Choose dates to use history. Search settings offers the original city-average view, other definitions and cleanest-day/hour searches; see methodology.

Current search freshness is deliberately stricter than the general 48-hour source-staleness alert. Offline mode can still answer current searches only while the saved observations satisfy that six-hour comparison policy. No AI API or web search runs during a user request.

The named station catalog is a verified, bundled snapshot, not a promise of continuously updated street coverage. Do not relabel national-map placeholder zeros as live readings. Source checks are in

DATA_SOURCES.md.

Database upgrade

Schema 1 is backed up automatically under `var\backups\before-schema-2-*.sqlite3` before migration to schema 2. A failed backup stops the migration. Existing measurements and city mappings are retained; a station-metadata table and hourly-search index are added. Old archive headers without station metadata remain accepted; new archives carry that metadata.

Keep the backup and use matching application/schema versions when rolling back. Do not run an old application against a migrated database.

To stop and release files before making a ZIP:

```
.\Stop-CleanAir.cmd
```

The stop launcher requests a normal server shutdown through workspace-local control files and waits up to 20 seconds. It validates the server PID, executable and process creation time, so a reused PID is not mistaken for the application. If needed, it stops leftover Python, Prometheus and Ruff processes whose executables belong to this workspace. Unrelated Python installations and other applications are not stopped.

It then flushes SQLite's WAL, checks database integrity and verifies the important database, log, monitoring and runtime file locks are released. It does not delete the database or measurement history. Wait for it to finish before making your ZIP. It is safe to run when the application is already stopped.

The launcher uses built-in Windows PowerShell with a process-only execution policy; it does not change the machine's execution policy or override organization-enforced policies.

Ctrl+C in the start console remains supported. A still-pending source request may be interrupted on process exit; transactions and checkpoints preserve committed imports. Shutdown control files are generated in `var\run` and are excluded from the handover ZIP.

Health and monitoring

Surface	Interpretation
`/health/live`	The application responds.
`/health/ready`	200 once initial history exists; 503 before backfill completes. A later source outage does not erase readiness.
`/api/status`	Available dates, progress, latest attempt, data freshness, worker state and comparison scope.
`/metrics`	API counters/latency, collection health, worker state, source age, progress and local row counts.
Prometheus `/targets`	Scrape status for the application.

Surface	Interpretation
Prometheus `/alerts`	Application-down, failed-collection, stopped-worker and stale-source rules.

Prometheus retains seven days by default. Alert states are available locally. Email, paging and external Alertmanager delivery are **not configured**. Application JSON logs rotate at 5 MB with three backups. Prometheus console logs are rotated on the next startup when they exceed 10 MB.

Source outage or bad source data

1. Check `/api/status` and `var\logs\application.jsonl`.
2. Existing historical results should still work; the UI should show the failure.
3. Correct network/source conditions and wait for the next configured refresh.
4. To run a manual retry, stop the app first:

   ```
   .\runtime\python\python.exe -m clean_air sync
   ```

5. Use `sync --force` only when an explicit re-read is needed. It bypasses ETags, not validation, precedence or duplicate protection.

Never replace failed imports with empty data or mock observations. Malformed files fail explicitly; other files can still be processed. If the initial backfill cannot complete, the application remains live and explains why history is not ready.

Source data is stale

Collection success means the upstream service responded, not that its observations are fresh. A separate age metric tracks the latest source measurement, with a 48-hour warning threshold by default. Investigate reporting delay and dataset availability; do not reset timestamps to silence an alert.

No city is eligible

Inspect the excluded-city list. Try another period or explicitly choose PM10, which currently has broader Bulgarian coverage. Do not silently substitute it for PM2.5 or lower quality settings without recording the change.

Worker has stopped

Inspect logs and `/api/status`. Prometheus alerts when collection is enabled but the thread is no longer alive. Restart after investigating the exception.

Backups

```
.\runtime\python\python.exe -m clean_air backup
```

This uses SQLite's backup API, runs an integrity check and writes a timestamped snapshot under `var\backups`. A backup can be taken while the service is running. Keep independent copies according to your organization's policy; keeping both the database and its backup on one physical disk does not protect against disk loss.

To recover:

1. Stop the application.
2. Keep the existing database, its WAL and logs together for diagnosis.
3. Use a verified backup as `var\data\clean_air.sqlite3` in a clean extracted copy of the project; do not combine it with another database's WAL.
4. Run the doctor command, then start the application.

```
.\runtime\python\python.exe -m clean_air doctor
```

Do not attempt automatic destructive database repair. Integrity failures are reported explicitly.

Archiving and restoring

The worker archives older-than-36-month records after a successful refresh. To run it explicitly while the app is stopped:

```
.\runtime\python\python.exe -m clean_air archive
```

Each archive is gzip-compressed JSON Lines containing versioned metadata, observations and daily records. Row counts and SHA-256 are verified before deleting active copies. The `.sha256` sidecar must stay beside the archive.

To restore an archive, place it and its checksum in `var\archives`, then:

```
.\runtime\python\python.exe -m clean_air restore before-YYYY-MM-DD-NUMBER.jsonl.gz
```

Restoration verifies checksum/counts, remaps source file identifiers, respects source precedence, rebuilds daily values and checks foreign keys. It is transactional and pauses archiving for 24 hours so the restored history can be queried. A source-only empty installation still needs its normal initial import.

Handover and relocation

Run `Stop-CleanAir.cmd`, wait for completion, then run `Prepare-Handover.cmd`. A busy workspace lock causes an explicit failure instead of packaging files while a writer is active. The local handover archive includes a consistent database snapshot and `.git` history when present in the checkout, but not transient state. Public

GitHub release archives omit internal Git metadata. Generated monitoring paths are regenerated after extraction. Python uses relative module paths, not a copied virtual environment's absolute paths. Build-machine-specific pip console launchers are omitted; commands use the bundled interpreter's `-m` entry points. The native Ruff helper is retained.

Verify ZIP hashes with PowerShell:

```
Get-FileHash -Algorithm SHA256 '.\dist\CleanAirFinder-Windows-x64.zip'
```

Compare with its `.sha256` sidecar. Extract to a different local directory and run the doctor command before handing it over.